

Iris

SOEN 490 Capstone Project Proposal

Community Service, Goods Redistribution, and Decision-Support Platform

Team / GitHub repository: Iris

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Project Description

Iris is a community-service platform for people and families in need, developed with B.A.D.R., a Montreal nonprofit organization. Its primary use case is the redistribution of surplus food, reflecting B.A.D.R.'s food-bank activities, but the platform uses a general goods model so that other approved necessities - such as baby products, hygiene items, clothing, and household essentials - can be supported without redesigning the system. The objective is to connect verified community need with available resources while giving the nonprofit better tools to manage eligibility, scarcity, referrals, and impact.

1. Beneficiary intake, verification, and eligibility. Applicants complete a guided conversational intake and, when required, upload supporting documents. The system extracts structured information, checks completeness and inconsistencies, and prepares a reviewable beneficiary profile. Eligibility rules will be confirmed with B.A.D.R. and aligned with applicable public criteria and recognized low-income measures. Relevant factors may include household resources, family composition, dependants, and other B.A.D.R.-approved vulnerability criteria. The system assists with verification; final policy remains under nonprofit control.

2. Goods redistribution and reservations. Businesses and partner organizations publish available goods with category, quantity, pickup location, and availability period. Verified beneficiaries can search, reserve, and collect approved goods at no cost. Reservation state and inventory controls prevent the same limited resource from being allocated more than once.

3. Needs, prioritization, and matching. Beneficiaries and community organizations can register current needs. The platform matches incoming supply with those needs and sends notifications. When demand exceeds supply, prioritization follows transparent rules configured with B.A.D.R.; for example, a very low-income household with several dependants may receive higher priority than a household with greater financial capacity. Every priority result remains explainable and auditable.

4. Community-resource assistant. A multilingual conversational assistant helps users search both current goods and a maintained database of local community services. It can guide a newcomer toward relevant administrative support or help a family find appropriate low-cost or free programs. Responses are grounded in approved resource data rather than general model knowledge alone.

5. Oversight, community intelligence, and impact reporting. B.A.D.R. staff receive dashboards on households supported, goods redistributed, successful pickups, unmet needs, participating donors, estimated value redistributed, and geographic supply-demand patterns. If document verification, duplicate-account checks, or usage monitoring raises a material concern, the affected account or transaction is placed in a Review Required state. The user cannot continue the affected allocation until an authorized B.A.D.R. staff member reviews the evidence and approves, requests clarification, or closes the case. No permanent exclusion is made solely by an AI model.

Technical approach. All modules share one beneficiary profile and system of record. AI is used for conversational intake, document extraction, resource retrieval, and review assistance; deterministic services handle eligibility rules, priority rules, reservation consistency, and other decisions that must be reproducible and testable.

Committed scope and evolution. The team is committed to an end-to-end workflow covering onboarding and verification -> needs and goods -> matching and prioritization -> reservation/pickup -> community-resource assistance -> impact reporting. The depth of individual modules may be expanded, simplified, or reprioritized as the project evolves with stakeholder feedback, available data, and technical findings, while preserving this core workflow. Possible extensions include person-to-person furniture redistribution, advanced anomaly detection, supply/demand forecasting, opportunity heatmaps, and route optimization.

Novelty, Innovation, and Disruption

- **A community-service platform, not simply a marketplace.** Iris is designed around verified people in need and nonprofit operations. Distribution, eligibility, prioritization, referrals, and reporting are part of one service workflow.
- **Demand-aware redistribution.** The platform models both available supply and expressed community need, allowing incoming goods to be matched to actual shortages instead of relying only on users browsing listings.
- **Transparent prioritization when resources are scarce.** Priority rules are defined with B.A.D.R., applied deterministically, and recorded so staff can explain why a household was prioritized.
- **Document-assisted eligibility.** Conversational intake and document extraction reduce repetitive administrative work while preserving human oversight and a traceable source for extracted information.
- **Human review before sensitive continuation.** Material verification or abuse concerns place the relevant workflow into review rather than allowing an automated model to make a final exclusion decision.
- **Grounded multilingual resource navigation.** The assistant searches live platform inventory and a curated community-resource knowledge base rather than acting as a generic chatbot.
- **Operational intelligence for the nonprofit.** The same system that redistributes goods also produces structured evidence of unmet needs, donor participation, successful redistribution, and geographic service gaps.
- **Extensible goods model.** Food is the main and initial use case, but the data model supports other approved necessities without creating a separate application for each category.

Core System Structure

Experience layer	Beneficiary, business/partner, and B.A.D.R. staff interfaces; multilingual conversational access.
Application services	Profiles, document review, donations, needs, matching, reservations, pickups, notifications, referrals.
Decision services	Eligibility rules, priority rules, allocation constraints, review status, and audit trail.
AI services	Conversational intake, document extraction, resource retrieval, discrepancy/anomaly assistance, narrative insights.
Data layer	Shared relational system of record, document metadata, community-resource knowledge base, and impact events.

Risk

- **Sensitive beneficiary information.** The platform may process personal or financial documents. Development will use synthetic or anonymized data whenever possible, minimize stored information, enforce role-based access, and restrict sensitive records to authorized users.
- **Incorrect document extraction or eligibility analysis.** Extracted values remain traceable to their source; conflicting or low-confidence results are flagged; deterministic eligibility rules remain separate from language-model output; and staff review is available for consequential cases.
- **Fairness when goods are scarce.** Criteria will be confirmed with B.A.D.R. and informed by applicable public eligibility/low-income frameworks. Rules are configurable and auditable, while staff retain authority for exceptional circumstances.
- **False abuse or duplicate-account suspicion.** A flag does not equal a finding of abuse. Affected actions enter Review Required status and can proceed only after authorized staff review and approval or clarification.
- **Safety and suitability of goods.** Supported categories, required listing details, availability windows, reporting/removal tools, and prohibited-item rules will be defined with the stakeholder.
- **Outdated community-service information.** The assistant retrieves from a maintained resource database, and authorized staff can update, verify, or remove records when programs change.

Competition

Search terms: free food sharing app; surplus goods redistribution Canada; community donation platform; food waste marketplace Canada; community resources Montreal; social services directory Quebec

Competitor 1 - OLIO

OLIO supports local sharing of free food and non-food items and lets users give away goods to others nearby. It is the closest competitor to the community-sharing side of Iris. Iris remains novel because it is built around a nonprofit-managed assistance workflow: beneficiary verification, structured needs, transparent priority rules, allocation, community-service referrals, and impact reporting are integrated into the same system rather than operating as general peer-to-peer sharing.

Competitor 2 - Too Good To Go

Too Good To Go connects consumers with stores and restaurants that have surplus food. Users reserve discounted surplus food and pay through the application before pickup. Iris addresses a different use case: its main food-redistribution workflow is charitable and free to eligible beneficiaries. It also models need, applies nonprofit-defined priority rules when resources are scarce, matches supply with demand, and provides B.A.D.R. with operational information about shortages and impact.

Competitor 3 - 211 Quebec

211 Quebec provides information and referral services for community, public, and parapublic resources, including a Greater Montreal service database. It overlaps with the resource-discovery component of Iris. Iris does not aim to replace 211; it connects resource discovery to an active assistance workflow in which a verified user can express a need, be matched with available goods, receive notifications, reserve resources, and interact with the nonprofit through the same platform.

Description of Customer and Company

Company

B.A.D.R. (Bureau Associatif pour la Diversite et la Reinsertion) is a Montreal nonprofit organization dedicated to fighting poverty and social exclusion and supporting vulnerable populations. Its community services include a food bank, furniture and clothing donations, solidarity transportation, school-supply support, education, sports and recreation, and community research/evaluation. [B.A.D.R. website](#)

Customer's Expertise

B.A.D.R. works directly with low-income individuals and families and already manages material and community assistance. Its current food-bank service specifically supports low-income families and individuals, and the organization also provides furniture/clothing assistance and other community programs. This gives the stakeholder practical expertise in beneficiary needs, intake, distribution workflows, local services, partnerships, and impact measurement.

Why the Customer Is Interested

B.A.D.R.'s original project brief proposes a digital solution connecting shops, grocery stores, supermarkets, and other businesses with surplus goods to individuals and families in vulnerable situations. The goal is to transform usable goods that could otherwise be discarded into free community support while respecting people's dignity. The brief also identifies university collaboration as an opportunity to validate user needs, study existing solutions, develop and test a prototype, address privacy and eligibility, establish partnerships, and measure social and environmental impact. Iris preserves that mission while adding structured needs, verified eligibility, transparent prioritization, automated matching, community-resource guidance, and decision-support analytics.

Eligibility and Priority Policy Basis

There is not one universal government definition of a 'family in need' that applies to every community program. For the project, eligibility and priority rules will therefore be co-designed and formally confirmed with B.A.D.R., then aligned with applicable Quebec program criteria and recognized official low-income measures. Quebec social-assistance rules consider resources, income, family composition, and other circumstances, while Statistics Canada's Market Basket Measure publishes low-income thresholds by region and family size. These public frameworks can inform the rule set, but B.A.D.R. remains responsible for approving the policy used by its own program.

References

- Stakeholder brief: Solidarity-Based Redistribution of Surplus Goods (B.A.D.R., 2026).
- [B.A.D.R. - Services and activities](#)
- [B.A.D.R. - About](#)
- [Government of Quebec - Social Assistance and Social Solidarity Programs](#)
- [Government of Quebec - Applying for Financial Assistance / Eligibility](#)
- [Statistics Canada - Market Basket Measure thresholds by region and family size](#)
- [OLIO](#)
- [Too Good To Go Canada](#)
- [211 Quebec](#)